

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 June 2026
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Project Name	ShopEZ: E-Commerce Application
Maximum Marks	4 Marks

Technical Architecture for ShopEZ:

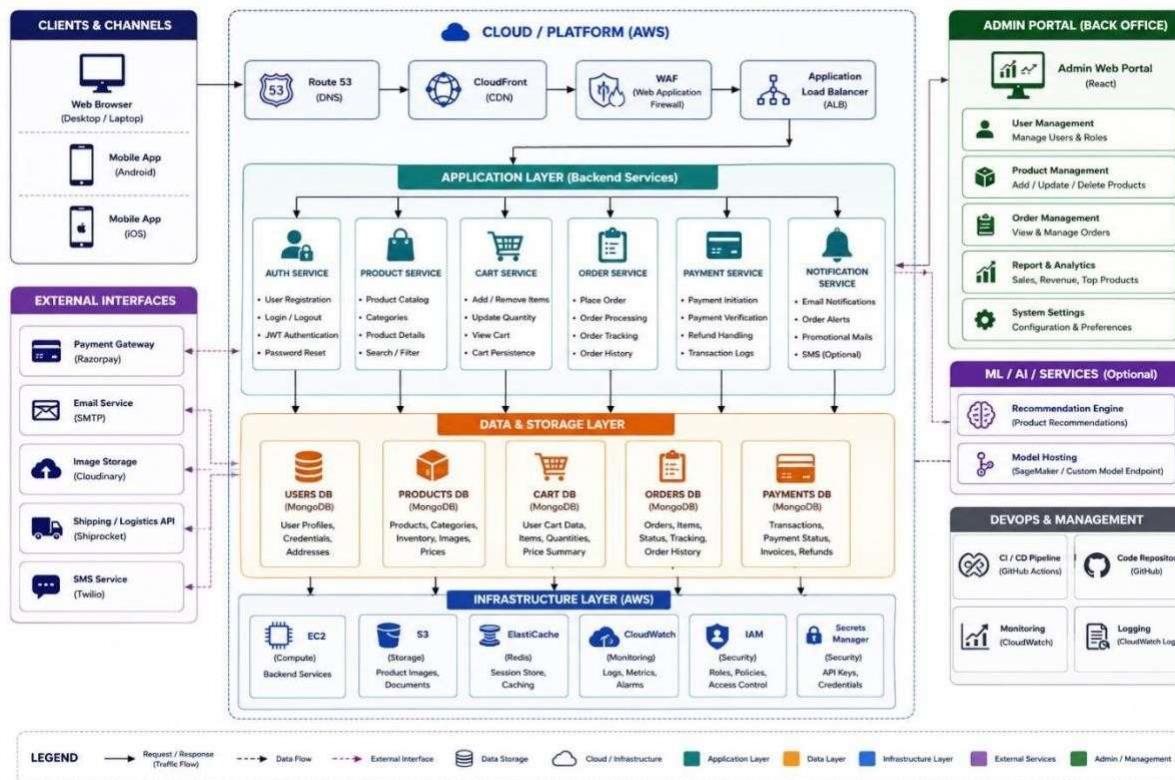


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Enables users to browse products and place orders.	React.js, Vite, Tailwind CSS
2.	Application Logic-1	Processes user authentication, cart, and order functions.	Node.js, Express.js, JWT, bcryptjs

3.	Application Logic-2	Handles product creation, updates, and retrieval.	Node.js, Express.js, Mongoose
4.	Application Logic-3	Provides tools for managing products and orders.	Node.js, Express.js, MVC Architecture
5.	Database	Stores user, product, cart, and order information.	MongoDB (Mongoose ODM)
6.	Cloud Database	Maintains application data on the cloud.	MongoDB Atlas
7.	File Storage	Stores and serves product images.	URL-based Image Storage
8.	External API-1	No external services integrated.	N/A
9.	External API-2	No third-party APIs utilized.	N/A
10.	Machine Learning Model	Not required for this application.	N/A
11.	Infrastructure (Server / Cloud)	Supports deployment and backend operations.	Node.js Server, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
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1.	Open-Source Frameworks	Developed using open-source web technologies.	React.js, Node.js, Express.js
2.	Security Implementations	Protects user accounts and application data.	JWT Authentication, bcrypt
3.	Scalable Architecture	Separates frontend, backend, and database layers.	React.js, Express.js, MongoDB
4.	Availability	Uses cloud services for reliable access.	MongoDB Atlas, Node.js
5.	Performance	Ensures smooth communication between system components.	MongoDB, Express.js, REST APIs